

Discussion Unit: DSE Analytics

Other (sub-servers)

- DSE Analytics History Server
- DSE Analytics Spark Job Server

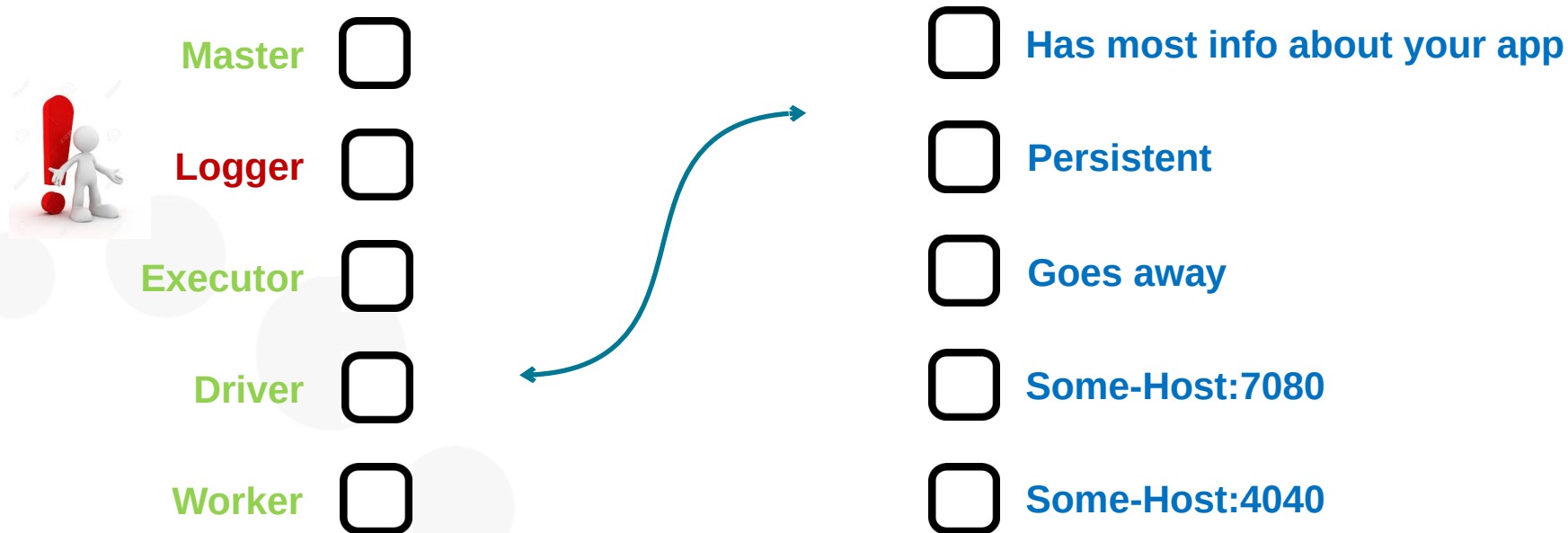
Not: DSE Always-on SQL (another unit)

Discussion Lab:

Matching pairs – Match the attributes on the right with the areas on the left

DSE Analytics; Process Architecture, and reporting on same

DSE Analytics: Process Architecture



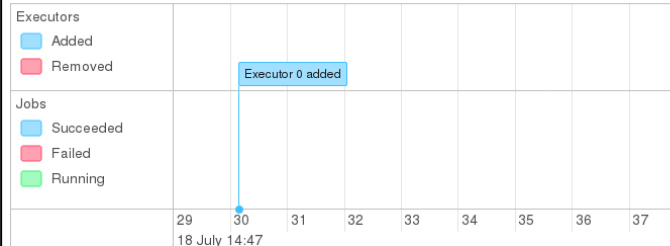
End of Discussion Lab:

While a DSE Analytics App is Running:

Spark Jobs (?)

User: root
Total Uptime: 17 s
Scheduling Mode: FIFO
Active Jobs: 1

Event Timeline
☐ Enable zooming



Active Jobs (1)

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks: Succeeded/Total
0	collect at App.scala:88 (kill)	2018/07/18 14:47:40	1 s	0/1	0/7

Stages for All Jobs

Active Stages: 1
Completed Stages: 1

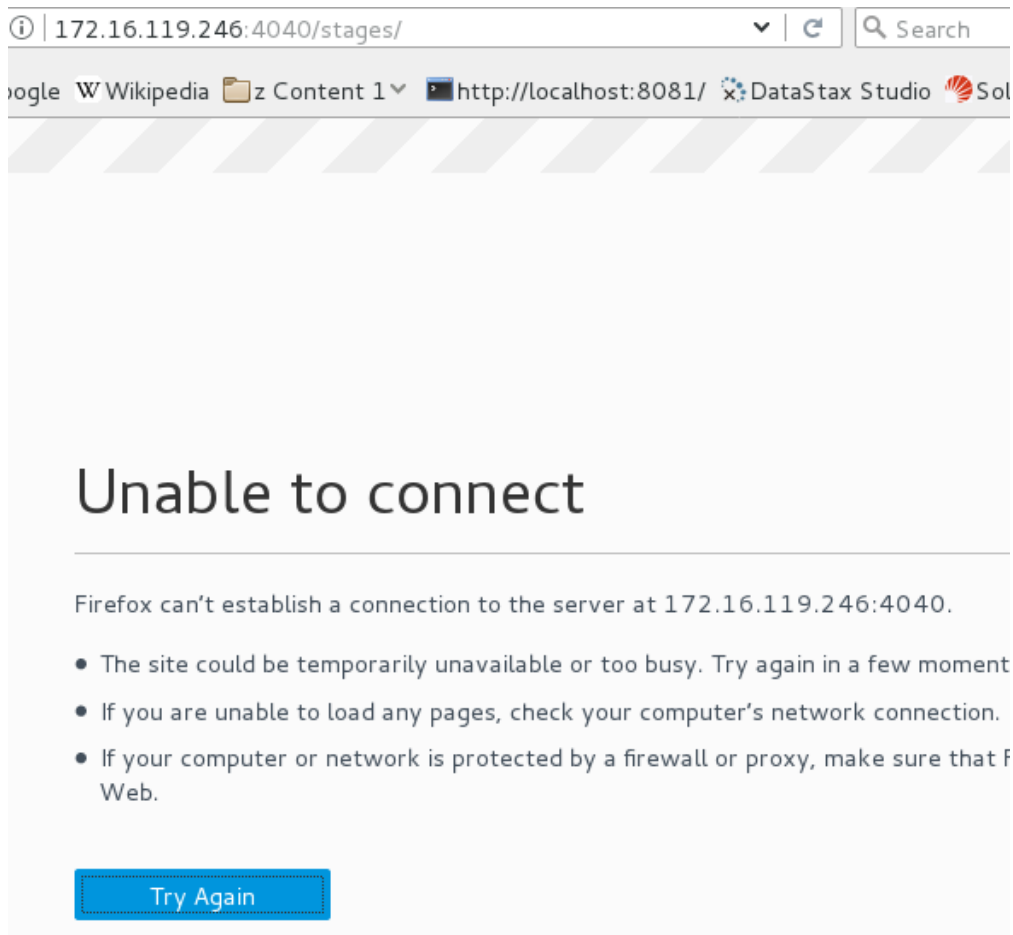
Active Stages (1)

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input
1	collect at App.scala:98 (kill) +details	2018/07/18 14:44:31	1 s	0/2	

Completed Stages (1)

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input
0	collect at App.scala:88 +details	2018/07/18 14:44:23	7 s	7/7	5.0 B

After a DSE Analytics App is Complete:



Spark Master at spark://127.0.0.1:7077 - Mozilla Firefox

File Edit View History Bookmarks Tools Help

New Tab x History Server x Spark Master at spark://... x +

localhost:7080

Google W Wikipedia Content 1 http://localhost:8081/ DataStax Studio Solr Admin Spark Master at spark://...

Spark Master at spark://127.0.0.1:7077

URL: spark://127.0.0.1:7077

Alive Workers: 1

Total Cores: 3 / 3

- default: 3 / 3

Total Memory: 1024.0 MB / 3.3 GB

- default: 1024.0 MB / 3.3 GB

Applications: 1 Running, 0 Completed

Drivers: 0 Running, 0 Completed

Status: ALIVE

Workers

Worker Id	Address	State	Cores
worker-20180719215834-127.0.0.1-39111	127.0.0.1:39111	DEAD	Total: 0 / 3 • default: 0 / 3
worker-20180719220055-127.0.0.1-38206	127.0.0.1:38206	ALIVE	Total: 3 / 3 • default: 3 / 3

Running Applications

Application ID	Name	Workpool	Cores	Memory per Executor	Submitted Time
app-20180719220355-0000 (kill)	My App	default	3	1024.0 MB	2018/07/19 22:03:55

Completed Applications

Application ID	Name	Workpool	Cores	Memory per Executor	Submitted Time	User	State	Duration
----------------	------	----------	-------	---------------------	----------------	------	-------	----------

After a DSE Analytics App is Complete:

Running Applications

Application ID	Name	Workpool
app-20180719220355-0000 (kill)	My App	default

Running Applications

Application ID	Name	Workpool

Completed Applications

Application ID	Name	Workpool	Cores
app-20180719220355-0000	My App	default	0



Executor Summary

ExecutorID	Worker	Cores	Memory	State
0	worker-20180719220055-127.0.0.1-38206	3	1024	EXITED

Running Executors (0)

ExecutorID	Workpool	Cores	State	Memory
------------	----------	-------	-------	--------

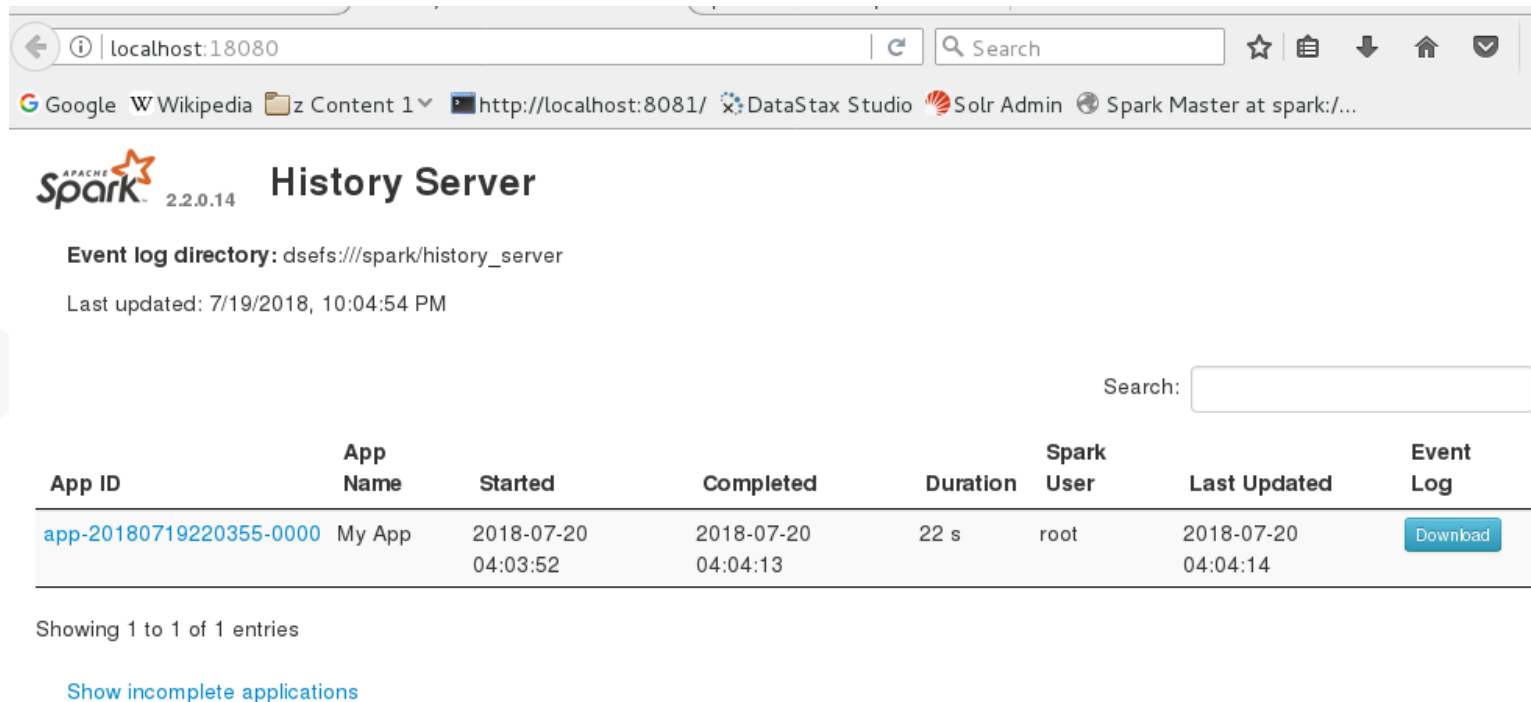


Finished Executors (1)

ExecutorID	Workpool	Cores	State	Memory	Job Details
0	default	3	EXITED	1024.0 MB	ID: app-201807 Name: My App User: anonymo

After a
DSE
Analytics
App is
Complete:

DSE Analytics History Server:



The screenshot shows a web browser window with the address bar at `localhost:18080`. The browser's tab bar shows several open tabs: Google, Wikipedia, a folder icon labeled 'z Content 1', `http://localhost:8081/`, DataStax Studio, Solr Admin, and Spark Master at `spark:/...`. The main content area of the browser displays the 'History Server' page. At the top left is the Apache Spark logo with version '2.2.0.14'. The title 'History Server' is prominently displayed. Below the title, it states 'Event log directory: dsefs:///spark/history_server' and 'Last updated: 7/19/2018, 10:04:54 PM'. A search bar is located on the right side of the page. Below the search bar is a table with the following columns: App ID, App Name, Started, Completed, Duration, Spark User, Last Updated, and Event Log. There is one entry in the table with the App ID 'app-20180719220355-0000', App Name 'My App', Started '2018-07-20 04:03:52', Completed '2018-07-20 04:04:13', Duration '22 s', Spark User 'root', and Last Updated '2018-07-20 04:04:14'. A 'Download' button is present in the Event Log column for this entry. Below the table, it says 'Showing 1 to 1 of 1 entries' and a link 'Show incomplete applications'.

History Server

Event log directory: dsefs:///spark/history_server

Last updated: 7/19/2018, 10:04:54 PM

Search:

App ID	App Name	Started	Completed	Duration	Spark User	Last Updated	Event Log
app-20180719220355-0000	My App	2018-07-20 04:03:52	2018-07-20 04:04:13	22 s	root	2018-07-20 04:04:14	Download

Showing 1 to 1 of 1 entries

[Show incomplete applications](#)

Spark Jobs (?)

User: root

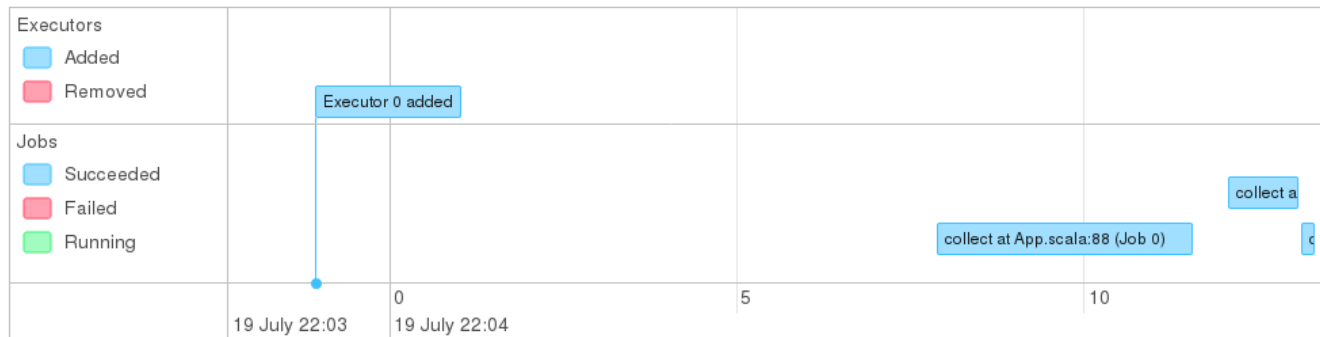
Total Uptime: 22 s

Scheduling Mode: FIFO

Completed Jobs: 3

▼ Event Timeline

☐ Enable zooming



Completed Jobs (3)

Job Id ▼	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	collect at App.scala:109	2018/07/19 22:04:13	0.2 s	1/1	2/2
1	collect at App.scala:98	2018/07/19 22:04:12	1 s	1/1	2/2
0	collect at App.scala:88	2018/07/19 22:04:07	4 s	1/1	7/7

DSE
Analytics
History
Server:

DSE Analytics History Server:

Environment

Runtime Information

Name	Value
Java Home	/opt/jdk/jre
Java Version	1.8.0_172 (Oracle Corporation)
Scala Version	version 2.11.8

Spark Properties

Name	Value
spark.hadoop.fs.ftp.host.port	21
spark.hadoop.fs.s3n.multipart.uploads.block.size	67108864
spark.hadoop.fs.s3n.multipart.uploads.enabled	false
spark.cassandra.connection.factory	com.datastax.bdp.spark.Ds
spark.sql.hive.metastore.barrierPrefixes	org.apache.spark.sql.cassa
spark.hadoop.fs.permissions.umask-mode	022
spark.hadoop.dse.client.configuration.impl	com.datastax.bdp.transport
spark.hadoop.hadoop.security.groups.cache.secs	300
spark.hadoop.fs.s3a.connection.ssl.enabled	true
spark.hadoop.hadoop.security.kms.client.encrypted.key.cache.expiry	432000000

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Blacklisted
Active(2)	0	0.0 B / 768.2 MB	0.0 B	3	0	0	11	11	12 s (0.4 s)	7.7 KB	0.0 B	0.0 B	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	0
Total(2)	0	0.0 B / 768.2 MB	0.0 B	3	0	0	11	11	12 s (0.4 s)	7.7 KB	0.0 B	0.0 B	0

DSE
Analytics
History
Server:

Executors

Show

20

 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Logs
driver	172.16.119.246:45682	Active	0	0.0 B / 384.1 MB	0.0 B	0	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	
0	127.0.0.1:44504	Active	0	0.0 B / 384.1 MB	0.0 B	3	0	0	11	11	12 s (0.4 s)	7.7 KB	0.0 B	0.0 B	stdout stderr

DSE Analytics History Server: How to

DSEFS

- `mkdir -p spark/history_server`

spark-defaults.conf (40 lines, below are adds)

- `spark.eventLog.dir dsefs:///spark/history_server`
- `spark.eventLog.enabled true`
- `spark.history.fs.logDirectory dsefs:///spark/history_server`

`dse spark-history-server start|stop`
`localhost:18080`

DSE Analytics History Server: Maintenance

spark-defaults.conf (more adds)

- spark.history.fs.cleaner.enabled true
- spark.history.fs.cleaner.interval 1d
- spark.history.fs.cleaner.maxAge 7d



On DSEFS: (JSON formatted), avail from Web UI

```
dsefs dsefs://127.0.0.1:5598/spark/history_server/ > ls -l
```

Type	Permission	Owner	Group	Length	Modified	Name
file	rw-rwx---	none	none	98468	2018-07-19 22:04:14-0600	app-20180719220355-0000



DSE Analytics Spark Job Server

DSE Analytics Spark Job Server:

How you run (applications) now:

- One of the REPLs
- "dse spark-submit ... "

How OSS Spark also runs applications:

- Spark REST Server
- Deprecated in DSE 5.1

DSE Analytics Spark Job Server-

- REST, Curl (no Bash required)
- Faster/easier than spark-submit .. (Re-uses SparkContext)
- Developers only, no HA

DSE Analytics Spark Job Server:

- Simple Web UI, port 8090
- Submit jobs via Curl
- Query status, results via Curl
- Query job history via Curl
- Can cache/share RDDs across jobs

DSE Analytics Spark Job Server: Configuring

- `./resources/spark/spark-jobserver/
./dse.conf
./settings.sh`
- `dse spark-jobserver start|stop`
- Tutorial on GitHub, Url on Notes page



End of Unit: